

Semaphore examples

Three concurrent processes X, Y and Z execute three different code segments that access and update certain shared variables. Process X executes the P operation (i.e. wait) on semaphores a, b and c; process Y executes the P operation on semaphores b, c and d; process Z executes the P operation on semaphores c, d and a before entering the respective code segments. After completing the execution of its code segment, each process invokes the V operation (i.e. signal) on its three semaphores. All semaphores are binary semaphores initialized to one. Which one of the following represents a deadlock-free order of invoking the P operations by the processes?

1. X:P(a)P(b)P(c) , Y: P(b)P(c)P(d) , Z: P(c)P(d)P(a)
2. X:P(b)P(a)P(c) , Y: P(b)P(c)P(d) , Z: P(a)P(c)P(d)
3. X:P(b)P(a)P(c) , Y: P(c)P(b)P(d) , Z: P(a)P(c)P(d)
4. X:P(a)P(b)P(c) , Y: P(b)P(c)P(d) , Z: P(c)P(d)P(a)

Ans 2

A binary semaphore with initial value given as 0, the following P & V operations are preformed on it in sequence 10 P, 15 V, 2 V, 4 P, 16 P, 2 V, 8 V. After all these operations the value of Binary semaphore & the no. of P operations that will remain block is _____

A counting semaphore S is initialized to 10. Then, 6 P operations and 4 V operations are performed on S. What is the final value of S?

$$= 10 - 6 + 4$$

$$= 8$$

A counting semaphore S is initialized to 7. Then, 20 P operations and 15 V operations are performed on S. What is the final value of S?

$$= 7 - 20 + 15$$

$$= 2$$

Let S be the binary semaphore variable $S = 0$ initially. Assume that no blocked processes exist in the system. The following signal (V), wait (P) operations are performed. The number of blocked processes at the end are _____

4 P, 6 V, 9 P, 13 V, 14 P